11(a)	Any three points from: • (max) energy of emitted electrons depends on frequency • (max) energy of emitted electrons does not depend on intensity • rate of emission of electrons depends on intensity (at constant frequency) • existence of frequency below which no emission of electrons • instantaneous emission of electrons • increasing the frequency at constant intensity decreases the rate of emission of electrons	**B3**
11(b)(i)	photon energy $= hc / \lambda$	**C1**
	$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (380 \times 10^{-9})$ $(= 5.23 \times 10^{-19}\ \text{J})$	**C1**
	$= 3.3\ \text{eV}$	**A1**
11(b)(ii)	photon energy must be greater than work function (energy)	**B1**
	so sodium and calcium	**B1**
11(c)	$\lambda = h / p$	**C1**
	$p = (6.63 \times 10^{-34}) / (380 \times 10^{-9})$ $= 1.74 \times 10^{-27}\ \text{N s}$	**C1**
	force $= 1.74 \times 10^{-27} \times 7.6 \times 10^{14}$ $= 1.3 \times 10^{-12}\ \text{N}$	**A1**